



# AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

## TEST REPORT NO. 2013 - 156



**GRUNDFOS Submersible Pump  
Model: SQF 2.5-2**

## GRUNDFOS SQF 2.5-2 Submersible Pump

|                        |   |                                                                                         |
|------------------------|---|-----------------------------------------------------------------------------------------|
| Test Report No.        | : | 2012 - 156                                                                              |
| Date of Test           | : | November 14 & 18, 2013                                                                  |
| Test Valid Until       | : | November 18, 2018                                                                       |
| Brand                  | : | GRUNDFOS                                                                                |
| Model                  | : | SQF 2.5-2                                                                               |
| Serial No.             | : | No data                                                                                 |
| Type and Size          | : | Single stage, helical rotor submersible pump                                            |
| Manufacturer           | : | GRUNDFOS A/S DK-8850 Bjerringbro, Denmark (Manufactured in Mexico)                      |
| Test Requested by      | : | GRUNDFOS PUMPS (Philippines), INC.<br>5548 Osmeña Avenue, Brgy. San Isidro, Makati City |
| Machine Classification | : | Commercial                                                                              |

### SUMMARY

The GRUNDFOS SQF 2.5-2 submersible pump was tested for performance. It was tested at the laboratory using AC power supply. The pump was tested started from a minimum Total Head of 15.0 meters to the maximum it can attain of 127.7 meters.

The Maximum Total Head (H) was 127.7 m occurred at 0.633 L/s Discharge Capacity. The Maximum Discharge Capacity of 0.811 L/s occurred at 15.0 m Total Head.

The Maximum Pump Efficiency of 58.1% was attained at a Discharge Rate of 0.763 L/s. At this optimum discharge condition, the Total Head was 85.4 m and the Input Power to the Pump shaft was 1.10 kW.

The Maximum Input Power to the Pump shaft of 1.45 kW occurred at the point of Maximum Discharge Capacity.

### PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance using electric power supply. Performance indicators included the Total Head, Input to the motor, and Pump Efficiency at varying discharge capacities. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on November 14 & 18, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. GRUNDFOS SQF 2.5-2 Submersible pump on the test set-up

### METHOD OF TEST

The pump was tested following the “ Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

### DESCRIPTION OF THE PUMP

The GRUNDFOS SQF 2.5-2 submersible pump, as shown in **Figure 2**, was a 32 mm single stage, helical rotor submersible pump. The suction inlet was immersed in water made of rings of small holes on the body of the pump. The discharge outlet were threaded and designed for a 32 mm diameter pipe. The motor has a maximum input power of 1400 watts was installed at the bottom of the unit. It was housed in a stainless body of the pump. The supply voltage can be 30-300 VDC or 90-240 VAC. It was controlled using a GRUNDFOS controller model: CU 200 SQFlex. It has also a safety device installed on the input wire which depending on water level will stop the operation of the pump if water fall below the pre-determined level of submergence of the safety device or will not operate if the water level is below the pre-determined level.

**Table 1** shows the detailed specifications of the pump.



Figure 2. GRUNDFOS SQF 2.5-2 Submersible Pump

### PUMP SPECIFICATIONS

Table 1. Detailed specifications of the GRUNDFOS SQF 2.5-2 Submersible Pump

|      |                              |   |                                               |
|------|------------------------------|---|-----------------------------------------------|
| 1.   | Brand                        | : | GRUNDFOS                                      |
| 2.   | Model                        | : | SQF 2.5-2                                     |
| 3.   | Serial No.                   | : | 95027414                                      |
| 4.   | Make                         | : | Mexico                                        |
| 5.   | Type                         | : | Single stage, helical rotor, submersible pump |
| 6.   | Size (mm)                    | : | 32                                            |
| 7.   | Overall Dimension and Weight |   |                                               |
| 7.1  | Length (mm)                  | : | 125                                           |
| 7.2  | Diameter (mm)                | : | 67                                            |
| 7.4  | Weight (kg)                  | : |                                               |
| 8.   | Impeller                     |   |                                               |
| 8.1  | Type                         | : | Helical rotor                                 |
| 8.2  | Material                     | : | Stainless steel                               |
| 9.   | Discharge Outlet             |   |                                               |
| 10.1 | Type                         | : | Threaded                                      |
| 10.2 | Diameter (mm)                | : | 32                                            |
| 10.3 | Material                     | : | Stainless                                     |
| 10.  | Casing                       |   |                                               |
| 11.1 | Type                         | : | Cylindrical                                   |
| 11.2 | Material                     | : | Stainless pipe                                |

### PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

|                                |   |                        |
|--------------------------------|---|------------------------|
| Pump Tested                    | : | GRUNDFOS               |
| Model                          | : | SQF 2.5-2              |
| Date of Test                   | : | November 14 & 18, 2013 |
| Test Condition                 |   |                        |
| Ambient Temperature (°C)       | : | 28.4                   |
| Atmospheric Pressure (mb)      | : | 1013                   |
| Pumping Water Temperature (°C) | : | 28.0                   |

Table 2. Results of Performance Test at 2000 rpm

| INPUT TO THE MOTOR |             |             | PUMP           |                          |                   |                     |
|--------------------|-------------|-------------|----------------|--------------------------|-------------------|---------------------|
| Power (kW)         | Voltage (V) | Current (A) | Total Head (m) | Discharge Capacity (L/s) | Output Power (kW) | Pump Efficiency (%) |
| 0.42               | 215.5       | 1.93        | 15.0           | 0.811                    | 0.119             | 28.4                |
| 0.50               | 215.1       | 2.47        | 29.1           | 0.810                    | 0.231             | 46.2                |
| 0.66               | 214.7       | 3.07        | 43.2           | 0.793                    | 0.336             | 50.9                |
| 0.79               | 214.2       | 3.67        | 57.3           | 0.788                    | 0.443             | 56.0                |
| 0.94               | 214.1       | 4.40        | 71.3           | 0.775                    | 0.542             | 57.6                |
| 1.10               | 214.1       | 5.10        | 85.4           | 0.763                    | 0.639             | 58.1                |
| 1.26               | 213.2       | 5.87        | 99.5           | 0.744                    | 0.726             | 57.6                |
| 1.42               | 212.9       | 6.77        | 113.6          | 0.722                    | 0.804             | 56.6                |
| 1.45               | 212.9       | 6.77        | 127.7          | 0.633                    | 0.792             | 54.7                |

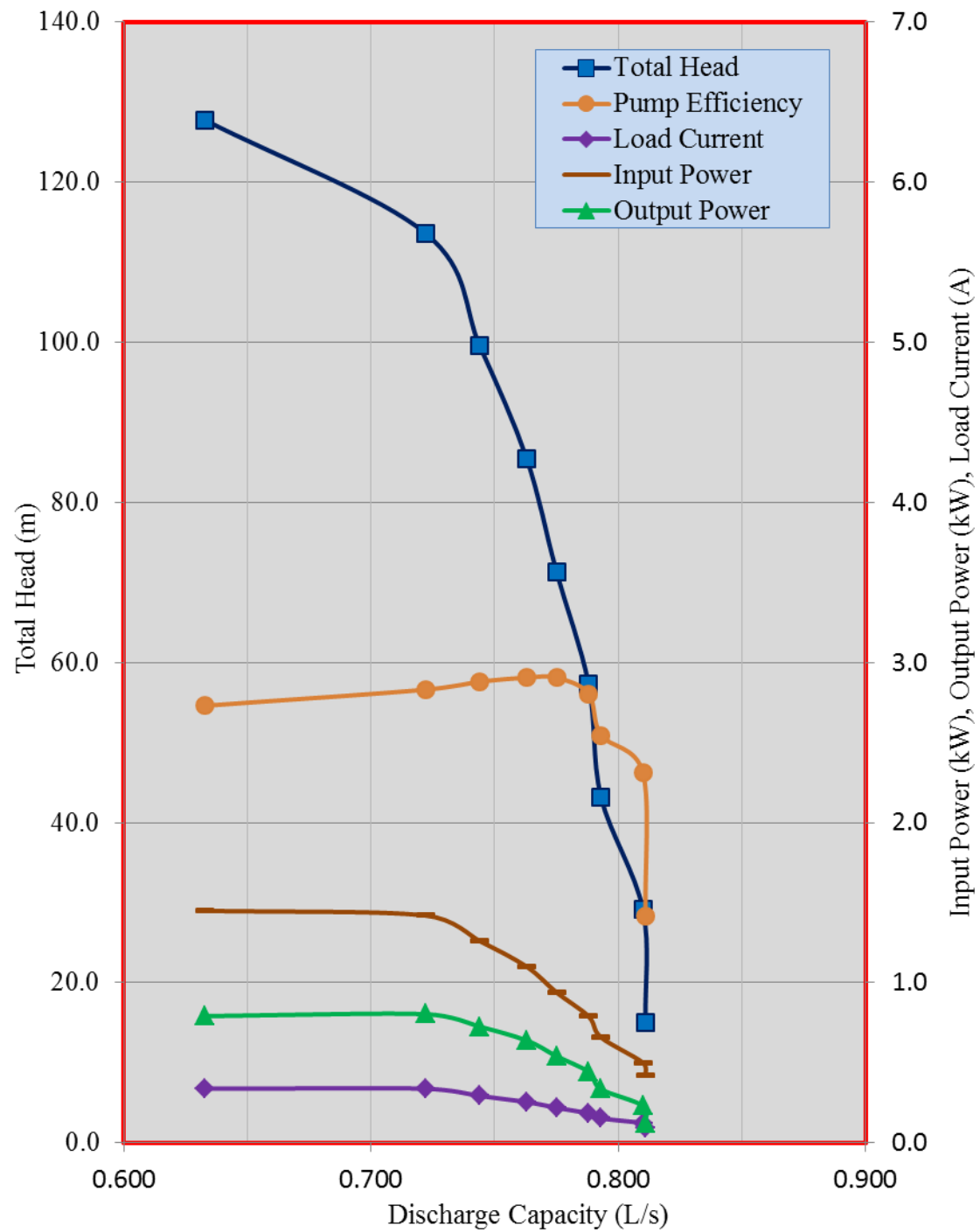


Figure 2. Performance Curve of the GRUNDFOS SQF 2.5-2 Submersible Pump

### DATA COMPUTATION

#### ANNEX 1. Formula used during calculation and testing

##### 1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

$H_d$  = Total Discharge Head (m)

$H_s$  = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$  = Pressure gage reading on the discharge pipe at  
gage connection converted to meter

$\frac{P_1}{\gamma}$  = Vacuum gage reading on the suction pipe at the  
point of gage connection converted to meter

$\frac{V_2^2}{2g}$  = Velocity head on the discharge pipe converted  
to meter. Computed by determining the velocity  
of liquid on the pipe by formula,  $V = Q/A$  where  
 $V$ , velocity of liquid inside the pipe in meter per  
per second,  $Q$ , Discharge rate of the pump in  
liters per second, and  $A$ , Cross-sectional area  
of the pipe in square meter.

$\frac{V_1^2}{2g}$  = Velocity head on the suction pipe converted to  
meter

## Annex (Cont'n)

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$Z_2$  = The distance from the center of the pressure gage face to the centerline of the pump in meters

$Z_1$  = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity,  $Q$ , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch
3. Water Power,  $WP$ , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power,  $IP$ , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

$T$  = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

$N$  = Input shaft angular speed in RPM

5. Pump Efficiency,  $\eta$ ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

$P_{atm}$  = Atmospheric pressure during the test converted to meters

$P_{vp}$  = Vapor pressure of the test liquid at certain temperature converted to meters

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Test Engineer:

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Approved for release:

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Date

N.B.

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